Quality Assurance Test Plan: Construction Field Management App

1. Scope of Testing

The testing parameters are strictly delineated to focus analytical resources on the application's most critical pathways.

- **In Scope:** The primary areas subjected to rigorous validation include the Authentication module (login, registration, password policies), the Project Management module (CRUD operations, status filtering, role-based access control), and the Daily Progress Report (DPR) module (submission workflows, date/worker count validation, and relational linkages).
- **Out of Scope:** Third-party payment gateway integrations, hardware-level performance profiling (e.g., thermal throttling, battery drain), and deep security penetration testing beyond basic authentication vulnerabilities are deferred to subsequent testing cycles.

2. Testing Types Used

To ensure comprehensive defect discovery, a multi-tiered testing methodology is deployed:

- **Functional Testing:** Verifies that core features like project creation and DPR submission execute flawlessly under normal conditions.
- **Negative Testing:** Intentionally provides invalid inputs (e.g., negative worker counts, unregistered emails) to validate error handling.
- **Boundary Value Analysis:** Tests the extreme edges of input fields, such as minimum password lengths and chronological limits.
- **UI/UX Testing:** Evaluates interface ergonomics, layout responsiveness, and readability under simulated field conditions (e.g., ensuring tap targets are large enough for users wearing gloves).
- **API Integration Testing:** Validates the backend RESTful architecture independent of the frontend to ensure correct status codes and payloads.
- **Regression Testing:** Employs automated scripts and baselines to verify that new code does not fracture previously stable functionalities.

3. Tools Used

Executing this test plan requires a specific suite of tools to accurately replicate the end-user experience and document results:

- **Test Case & Bug Documentation:** Spreadsheets (Google Sheets / Microsoft Excel) for centralized tracking.
- **API Testing:** Postman (utilizing JSON collections for automated endpoint validation).
- **UI & Automation Testing:** Cypress (JavaScript-based E2E framework) and modern web browsers (Chrome, Firefox, and mobile viewports via iOS Safari/Android Chrome simulation).

4. Entry and Exit Criteria

Strict quality gates govern the testing lifecycle to prevent premature deployment.

- **Entry Criteria:** Testing execution formally commences only when the development phase for the target modules is certified complete, the staging environment is stable with isolated databases, and comprehensive API specifications are available.
- **Exit Criteria:** The testing phase is authorized for exit only when 100% of the defined test cases have been executed and documented. Furthermore, no defects classified as 'Critical' or 'Major' severity may remain unresolved, and the overall software defect density must fall within acceptable enterprise benchmarks.